

AVNEET KAUR

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EXPERIENCE

RAC-DRDO

Jun 2025 – Present

Project Intern (Recruitment and Assessment Centre, DRDO)

- Designed the architecture for a Gen AI-powered Interview Panel Curation system, leveraging ML similarity measures and LLMs.
- Developed the system to reduce manual effort and improve match accuracy by **60%**.
- Integrated web scraping to auto-fetch evaluator profiles, enabling real-time panel recommendations.

PROJECTS

Pravah: AI-Powered Smart Traffic Management System | SIH'25 Winner

Sep 2025 – Present

- Tech: Python, OpenCV, SUMO, Pygame, RL (PPO), LSTM, FastAPI
- Architected a three-tier closed-loop system using **PPO Reinforcement Learning** to dynamically optimize signal phases, increasing operational efficiency by **50%** compared to legacy fixed-timers.
- Engineered a **YOLOv8-based** computer vision pipeline classifying **6+ vehicle types** in real-time, integrated an emission-weighted priority scoring engine with zone-aware multipliers for sensitive areas (schools, hospitals) to pre-empt red signals, prioritizing emergency vehicle throughput by **26%** and reducing high-emission vehicle idle time by an estimated **30%**.
- Developed a **FastAPI**-driven commuter app using **LSTM** forecasting to predict gridlock 20 minutes ahead, achieving a **15–20%** reduction in peak-hour congestion.
- Implemented a multi-objective reward function achieving a **10.5%** reduction in CO₂ emissions by creating “green waves.”

MIRA: Matching Interviewers with Resume using AI | SIH'24 Winner | [GitHub](#)

Sep 2024 – Dec 2024

- Tech: Mistral-7B, SBERT, BERT, Vector DB, ETL Pipelines
- Built an **LLM-based** expert panel matcher that curates interviewers by skill-expertise alignment, reducing HR effort and manual screening time.
- Implemented a resume parsing pipeline with **ETL** and vector embeddings, cutting processing time by **40%** through automated data handling and semantic matching.
- Enabled low-latency contextual recommendations using **dense vector search** under high query loads.

Convera AI: Voice-Based Knowledge Builder for AI Agents | Hackaccino Winner | [GitHub](#)

Feb 2025 – Jun 2025

- Tech: Mistral (Ollama), Whisper, FastAPI, Web Speech API, Flask, SQLite
- Engineered a voice-driven pipeline converting conversations into structured **RAG** datasets, reducing manual data entry by **85%**.
- Integrated **OpenAI Whisper** for low-latency speech-to-text and LLM-based prompting to extract domain-specific context via voice-driven questioning.
- Optimized onboarding for non-technical users, increasing accessibility by **40%** via a conversational voice-call interface.

EDUCATION

Bennett University

2023 – Present

Bachelor of Technology (B.Tech) in Computer Science

Greater Noida, India

- CGPA: **8.18 / 10.0**

SKILLS

Languages: C++, Python, HTML, CSS, SQL (MySQL) **Tools & Frameworks:** Scikit-learn, Git, MongoDB
Core Concepts: AI/ML, LLM Integration, RAG, Reinforcement Learning (PPO), YOLO, LSTM, Generative AI

ACHIEVEMENTS & AWARDS

Smart India Hackathon 2025 – National Winner
Smart India Hackathon 2024 – National Winner
Agentic AI Hackathon 2025 – International Winner (Ulster University, UK)
Cyber AI Hackathon 2025 – International Winner (University of Derby, UK)
Student Innovation Excellence Award 2025 – Education Summit–North, Times Now
AI Innovate 2025 (Microsoft) – 1st Runner-up, Top 1.7% among 300+ teams
Hackaccino 2025 – Track Winner (1800+ participants)
2× Smart BU Hackathon ('24 & '25) – Top 1.75%, Ranked 15 & 8 out of 400+ teams